

CoARA Action Plan for SciLifeLab: A Path Towards Recognising Diverse Contributions to Research Through Open Science and the FAIR Principles

About SciLifeLab

SciLifeLab (www.scilifelab.se) is a hub for life science research in Sweden and a collaboration between Swedish universities. Everyone at SciLifeLab is employed at one of the partner universities. Since 2013, SciLifeLab has been assigned by the Swedish government to operate national infrastructure, to provide cutting-edge life science technologies and -expertise to Swedish researchers, and to be a top international centre for research in health- and environment sciences.

Introduction

SciLifeLab [announced](#) that it had joined CoARA on 23 February 2024 with the goals of advancing research assessment practices and open science within the organisation, while collaborating with the [CoARA Swedish National Chapter](#), CoARA community-wide groups and chapters, as well as global membership. A new [CoARA-SciLifeLab Working Group](#) will develop the steps the organisation will take to align with CoARA's objectives. The following plan documents the Working Group's responses to CoARA's instructions and guiding questions.

By adopting CoARA, SciLifeLab aims to drive scientific breakthroughs through open science approaches, such as enhanced data and software sharing, and to recognise diverse contributions, across the research lifecycle. This collaborative approach supports impactful, inclusive research that values all roles essential to scientific progress.

Guiding Principles

SciLifeLab's CoARA action plan aligns with its mission to enable collaborative, interdisciplinary life science research. Research is advanced through real-time data-sharing programmes and interdisciplinary platforms that ensure broad access to research infrastructure, support interoperability with bioinformatics and AI tools, and enhance data accessibility through unified systems with enriched metadata. We prioritise open science and FAIR principles to ensure research outputs—data, software, methodologies, publications, and training materials—are open, findable, accessible, interoperable, and reusable (FAIR). The plan is also tied to the [SciLifeLab Data Policy](#) which outlines guidelines for data sharing, management, and accessibility to promote

broad, reproducible, and ethical life science research, supporting national and international research initiatives.

SciLifeLab values and recognises diverse research contributions beyond traditional publications, celebrating team science and the collaborative efforts that drive advancements in life sciences. As a hub for FAIR-aligned training resources, SciLifeLab also promotes data reusability across diverse life sciences fields. Community engagement is central to our approach, fostering inclusive spaces where diverse voices are valued, supported by [SciLifeLab's Code of Conduct](#) and feedback mechanisms. We encourage our community to actively contribute to CoARA goals.

Working Group Formation and Objectives

In the spring of 2024, SciLifeLab formed a dedicated Working Group, bringing together experts from across the organisation to address and enhance research assessment practices in alignment with the CoARA initiative. Chaired by Chris Erdmann, Head of Open Science, and co-chaired by Katarina Öjefors Stark, Data Steward, from the SciLifeLab Data Centre, this multidisciplinary team comprises members including Wei Ouyang, Laura Carroll, Erdinc Sezgin, Annika Jenmalm Jensen, Jessica Lindvall, Niclas Jareborg, Ulrika Wallenquist, and Aristidis Moustakas, joined by members of the Open Science team, Suné Joubert, Parul Tewatia, and Joanna Sendecka. Ex-officio representatives from partner universities have been invited to ensure planning alignment and share their perspectives. Together, the Working Group members bring a wealth of knowledge to advance the CoARA objectives within SciLifeLab's context, and as work progresses, the Working group will also be open to iteratively furthering inclusive representation.

The Working Group will lead SciLifeLab's CoARA Action Plan, with a focus on key issues such as promoting data and software sharing and enhancing open science practices both within SciLifeLab and across the broader research community. Over the past decade, SciLifeLab has developed infrastructure and policies supporting open science and FAIR data management. However, limited incentives have hindered full researcher adoption, making CoARA engagement an essential step towards building a more supportive environment for open science principles.

Operating as a collaborative and inclusive forum, the Working Group plans to meet 2–4 times a year and will host 2–4 events, such as webinars, to explore core CoARA topics, FAIR principles, and SciLifeLab's guiding principles on open science. These events aim to provide valuable insights and to foster community engagement through open discussions. Acting as a conduit for feedback, the group will iteratively review and adapt its goals based on community input. The Working Group will identify and connect with other SciLifeLab groups addressing related aspects of CoARA. The Working Group will also serve as a sounding board, addressing questions and gathering feedback on CoARA-related topics from community members, including researchers and other stakeholders.

Communication and knowledge-sharing will be conducted through a variety of channels, including social media, email, Slack, newsletters, and webinars. Reports, training materials, and other resources will be shared openly via platforms like Zenodo and Figshare, supporting transparency and accessibility. This interactive, community-centred approach ensures that the

Working Group's activities remain flexible and responsive to feedback, strengthening SciLifeLab's commitment to fostering a welcoming and inclusive environment.

In taking these steps, the Working Group seeks to lead SciLifeLab's reform efforts, aligning with its guiding principles to build a more transparent, accessible, and collaborative research culture.

How We Will Work

To achieve CoARA goals, we will leverage available resources and focus on addressing key challenges related to the diversity of research outputs, engagement of fellows, and training initiatives. Recognising gaps and barriers within our organisation, particularly in reforming research assessment, we aim to enhance authority and resource allocation to effectively address these needs. Diversity in research contributions and career paths will be valued through multiple channels—community conversations, Working Group collaborations, and both qualitative and quantitative metrics. Our efforts will include qualitative insights gathered through community engagement, as well as formal research on assessment practices from both policy work and research collaborations.

We will link these initiatives to annual reporting and open community discussions that examine traditional evaluation forms alongside new, progressive assessment approaches. To support these, we plan to utilise in-kind time, departmental budgets, and existing financial structures, while exploring opportunities for additional funding. New and alternative forms of research communication, updates on research reform trends, and emerging practices will be shared with the community to ensure alignment with cutting-edge developments. Through Working Group members, we will engage with relevant groups, panels, and committees to promote CoARA objectives, opening dialogue about our progress and challenges. Monitoring mechanisms will also be put in place to track our advancements towards CoARA goals, allowing us to adjust and enhance our approach as needed.

Swedish National Chapter and CoARA Community

SciLifeLab has joined the Swedish National Chapter of CoARA, reinforcing its commitment to advancing research assessment practices in Sweden. The Swedish National Chapter, led by Jan-Ingvar Jönsson, Vice-Chancellor of Linköping University, plays a crucial role in supporting national and regional implementation of the Agreement on Reforming Research Assessment. The primary role of National Chapters is to facilitate mutual learning and discussions focused on research assessment, with opportunities to address broader topics, such as open science. Through its involvement in the Swedish National Chapter, SciLifeLab will benefit from shared insights and contribute to the broader Swedish scientific community.

Furthermore, SciLifeLab will work closely with partner universities to align efforts and ensure a cohesive approach, avoiding isolated action. By engaging with the wider CoARA community, SciLifeLab aims to gain valuable insights, contribute its own experiences, and strengthen its role in shaping sustainable, inclusive research practices.

Milestones

Across all years

- 2-4 Working Group meetings per year, 1-2 hour(s) each
- 2-4 CoARA-related SciLifeLab-wide events, such as webinars, per year
- Participation and collaboration in Swedish National Chapter, CoARA Working Groups
- Align and integrate CoARA-based open science and FAIR activities into SciLifeLab's data management services, training, and capacity building
- Integrate Training Hub activities by leveraging it as a platform to recognise and value training efforts, develop resources aligned with CoARA commitments, and drive cultural change

Year 1

- Join CoARA and the Swedish National Chapter
- Form Working Group and provide background on CoARA
- Draft action plan, review, and publish

Year 2

- Release of open science monitoring dashboards (e.g., demonstrating SciLifeLab data, software, etc sharing)
- Collect feedback and findings from monitoring to inform SciLifeLab's policies (e.g. Data Policy) and guidance (e.g., checklists)
- Develop and implement community initiatives to foster open science and FAIR (e.g. communities of practice)
- Launch open science web presence and resources for SciLifeLab

Years 3-5

- Revisit, review, and potentially revise plan including years 3-5 milestones
- Explore and pursue opportunities to incorporate open science and FAIR principles into recruitment, evaluation, promotion, and awards (e.g., learning from the Fellows Programmes to start, leveraging Voices for open science stories)
- Iteration and improvement of Year 2 initiatives and projects

List of Contributors to the Action Plan

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Repository

Working Group and community materials will be shared in the following Zenodo community:

<https://zenodo.org/communities/opensci-scilifelab-dc>

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